

Exercise 4

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The NHANES dataset is collected by the CDC and is a survey that assesses a participants (all ages) health and nutrition. This report focuses on the demographics of the data including Age, Gender, and Ethnicity and their relationship with Systolic Blood Pressure readings.

Preparing NHANES table

```
##
## 2015 2016 2017
## 10175 9971 9254

##
##      Female Male
## 2015  5172 5003
## 2016  5079 4892
## 2017  4697 4557

##
##      Female      Male
## 2015 50.83047 49.16953
## 2016 50.93772 49.06228
## 2017 50.75643 49.24357

## # A tibble: 3 x 4
##   wave_id 'Sample size' 'Males (n)' 'Male (%)'
##   <int>      <int>      <int>      <dbl>
## 1  2015         10175         5003        49.2
## 2  2016          9971         4892        49.1
## 3  2017          9254         4557        49.2
```

Number of Male Participants in NHANES Survey			
Wave	Total Sample size (n)	Male (n)	Male (%)
2015	10175	5003	49.2
2016	9971	4892	49.1
2017	9254	4557	49.2

Note:

Male (%) represents the percentage of males within each wave.

```
##
##      Asian Black Mexican Other Other Hispanic White
## 2015  1074 2267   1730  470          960 3674
## 2016  1042 2129   1921  505         1308 3066
## 2017  1168 2115   1367  634          820 3150
```

```
##
##           Asian      Black   Mexican      Other Other Hispanic      White
##  2015 10.555283 22.280098 17.002457 4.619165      9.434889 36.108108
##  2016 10.450306 21.351921 19.265871 5.064688      13.118042 30.749173
##  2017 12.621569 22.854982 14.771990 6.851091      8.861033 34.039334

## # A tibble: 3 x 3
##   wave_id 'Ethnicity (n)' Percentage of Sample Size[, "~1" [, "Black"] [, "Mexican"]
##   <int>      <int>      <dbl>      <dbl>      <dbl>
## 1  2015      10175      11      22      17
## 2  2016      9971      10      21      19
## 3  2017      9254      13      23      15
## # i abbreviated name: 1: 'Percentage of Sample Size'[, "Asian"]
## # i 1 more variable: 'Percentage of Sample Size'[4:6] <dbl>

## # A tibble: 3 x 6
##   wave_id Minimum Maximum Median Mean 'Standard deviation'
##   <int>    <dbl>    <dbl>  <dbl> <dbl>      <dbl>
## 1  2015     64.7     229.   115.  118.      18.1
## 2  2016     74      231.   117.  120.      18.4
## 3  2017     72.7     234    118   122.      20.3
```

Wave	Average systolic blood pressure (mmHg)				
	Minimum SBP	Maximum SBP	Median SBP	Mean SBP	
2015	64.66667	228.6667	115.3333	118.3159	
2016	74.00000	231.3333	117.3333	120.4020	
2017	72.66667	234.0000	118.0000	121.6625	

Note:

Values are reported in mmHg. SBP stands for systolic blood pressure reading. Summary statistics are based on the average